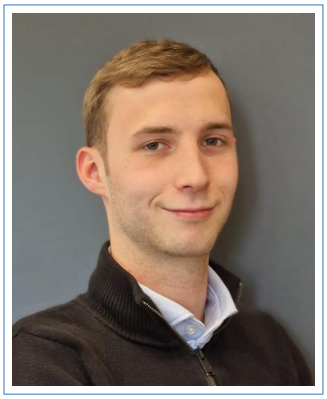
Aaron Kreis

M.Sc. Physics Candidate
Expected 2027

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Experimental Physics · Optical Imaging · Scientific Computing

Research Experience

06 – 08/2026 **Internship – Automated Bioluminescence Microscopy and Fourier Light-Field Imaging**, *NMSB Group – Prof. Michael Krieg — ICFO – The Institute of Photonic Sciences*

- Expanded and characterized a multimodal microscopy platform for brightfield, fluorescence, bioluminescence, and Fourier light-field microscopy (FLFM)
- Integrated a motorized filter wheel and designed/modified high-precision CAD models for optomechanical improvements to the microscope
- Developed Micro-Manager plugins for automated spiral sample scanning and configurable multi-step acquisition workflows controlling camera, XY/Z stages, illumination, and filter wheel
- Characterized the FLFM system using experimental point-spread-function measurements and reconstructed volumetric C. elegans data using Richardson–Lucy deconvolution; additionally evaluated deep-learning-based LUCID reconstruction models

07 – 10/2025 **Bachelor Thesis – Light Ray Propagation in Microscopic Imaging of Red Blood Cells**, *Dynamics of Fluids Group – Prof. Christian Wagner — Saarland University*

- Developed ray tracing simulations to model light propagation in microscopic imaging
- Simulated optical interactions with rotationally symmetric biological structures
- Performed quantitative intensity analysis on the detector plane
- Designed a computational workflow for reproducible microscopy simulations

Awards & Grants

2026 – 2027 **Glückauf Scholarship** – RAG-Stiftung
05/2026 **Santander Scholarship for Erasmus+ Mobility** — Santander Deutschland
03/2026 **Travel Grant – DPG Spring Meeting (SKM)** — WE Heraeus-Stiftung
2025 – 2026 **Glückauf Scholarship** – RAG-Stiftung
03/2025 **Travel Grant – DPG Spring Meeting (SKM)** — WE Heraeus-Stiftung

Technical Skills

Programming Python, MATLAB, C++
Engineering CAD, 3D printing
Tools LaTeX, Git, Linux

Education

2025 – 2027 (expected) **M.Sc. Physics (Double Degree)** — Université Grenoble Alpes & Saarland University
2021 – 2025 **B.Sc. Physics** — Saarland University
2013 – 2021 **AbiBac** — Warndt-Gymnasium

Teaching Experience

2023 – 2025 **Instructor (independently taught)** — Saarland University
- Summer School "From Atoms to Zettabytes" for STEM-oriented pre-university students
- Mathematics Preparatory Course for incoming undergraduate students

2022 – 2025 **Teaching Assistant for Undergraduate Physics Courses** — Saarland University
- Led problem-solving sessions accompanying undergraduate lectures
- Guided students in experimental methods, data analysis, and scientific reporting
- Supervised laboratory courses and tutorial sessions in mechanics, electromagnetism, and computational physics

Academic Service & Leadership

2022 – present **Student Representative** — Saarland University
Examination Board (B.Sc./M.Sc. Physics); Appointment Committees; Faculty Council; Student Council

2025 – 2026 **Vice President for Partnerships** – PhITEM Student Council — Université Grenoble Alpes

2024 – 2025 **Founding Deputy Chair** – jDPG Regional Group Saarbrücken — German Physical Society

2024 **External Review Panel Member** – B.Sc. Physics Program — University of Luxembourg

2023 – present **Member**, German Physical Society (DPG)

Employment

2022 – 2025 **Student Assistant – Physics Outreach** — Department of Physics – Saarland University
- Designed and delivered physics demonstrations for school visits, career fairs, and public events
- Represented the department in student recruitment initiatives

2021 – 2023 **Student Assistant – Process Automation** — DEKRA Automobil GmbH
- Developed automation workflows to streamline administrative processes across multiple departments
- Reduced manual workload by optimizing repetitive tasks

Languages

German Native

English $\geq$B2

French $\geq$B2